

Bike Station Availability & Utilization Analysis

1. Project Overview and Objective

This project analyses the operational performance of a bike-sharing network using an interactive Power BI dashboard. The objective of the analysis is to evaluate bike availability, station capacity, utilization efficiency, and geographic distribution of stations across multiple countries. By leveraging data visualization and analytics, the dashboard provides actionable insights that support operational optimization and strategic decision-making for bike-sharing services.

2. Data Sources

- **Source Description and Timeline:** Entri Elevate
- **Domain:** Transportation Analytics (Bike-Sharing / Urban Mobility)

3. Problem Statement

To analyse bike-sharing station availability, capacity, and utilization in order to identify operational inefficiencies and improve bike distribution, station planning, and overall network performance.

4. Attribute (Column /Features) Details:

Fact_Bike_Status(Fact Table)

Attribute Name	Data Type	Description
station id	Whole Number	Unique ID for station
bike stands	Whole Number	Total number of bike stands
available bike stands	Whole Number	Number of available bike stands
available bikes	Whole Number	Number of available bikes
date	Date	Date

Dim_station (dimension table)

Attribute Name	Data Type	Description
station_number	Whole Number	Unique ID for station
station_name	Text	Name of the station
latitude	Decimal	Latitude of the station
longitude	Decimal	Longitude of the station
city	Text	City where the station located
country	Text	Country where the station located
banking	Text	True/false
bonus	Text	True/false
status	Text	OPEN/CLOSED

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
 - Position column is separated by using comma in power query as latitude and longitude.
 - Last Update column is separated using left most space in order to get short date and renamed Last Update.1 to last update and Last Update.2 to last update time
 - Extracted only station name from name column using Transform tab→ Extract → Text After Delimiter '-' and then trimmed spaces.
 - Removed Blank station name
 - Capitalized Each word of Contract Name column and renamed it as city
 - <https://www.geoapify.com/tools/reverse-geocoding-online/> -- generated city and country with latitude and longitude values
 - Created fact_bike_status table and dim station table in excel.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals. **(List Out)**

Total Bikes = SUM(Fact_Bike_status[available bikes])

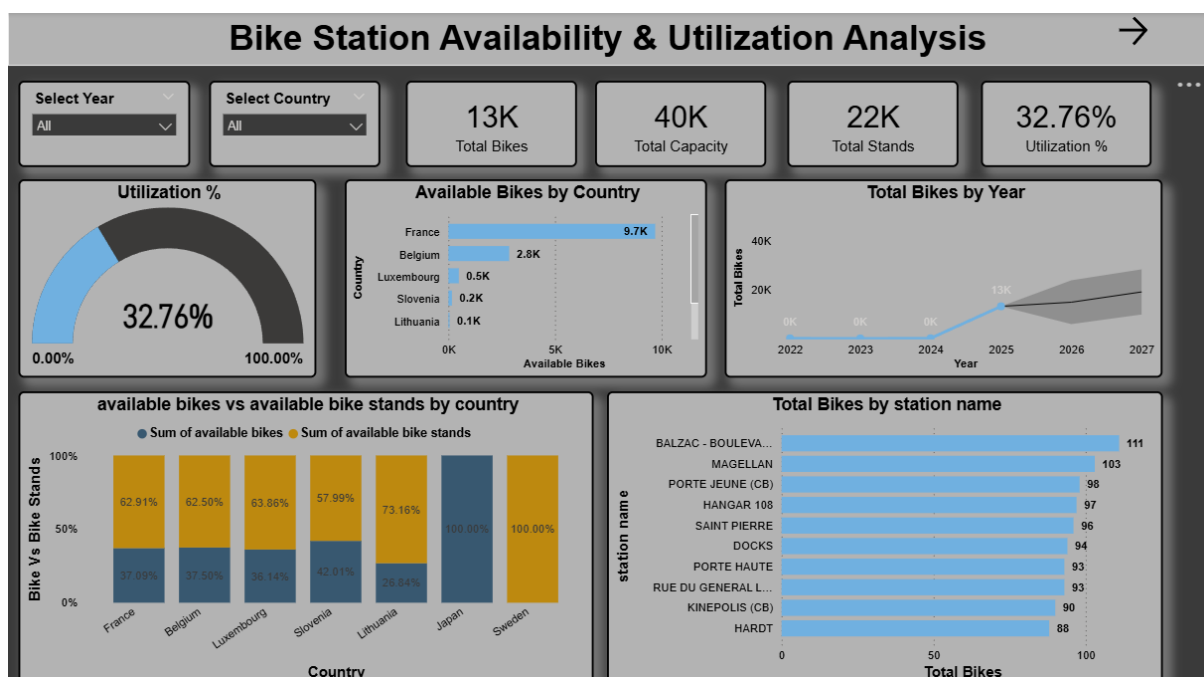
Total Stands = SUM(Fact_Bike_status[available bike stands])

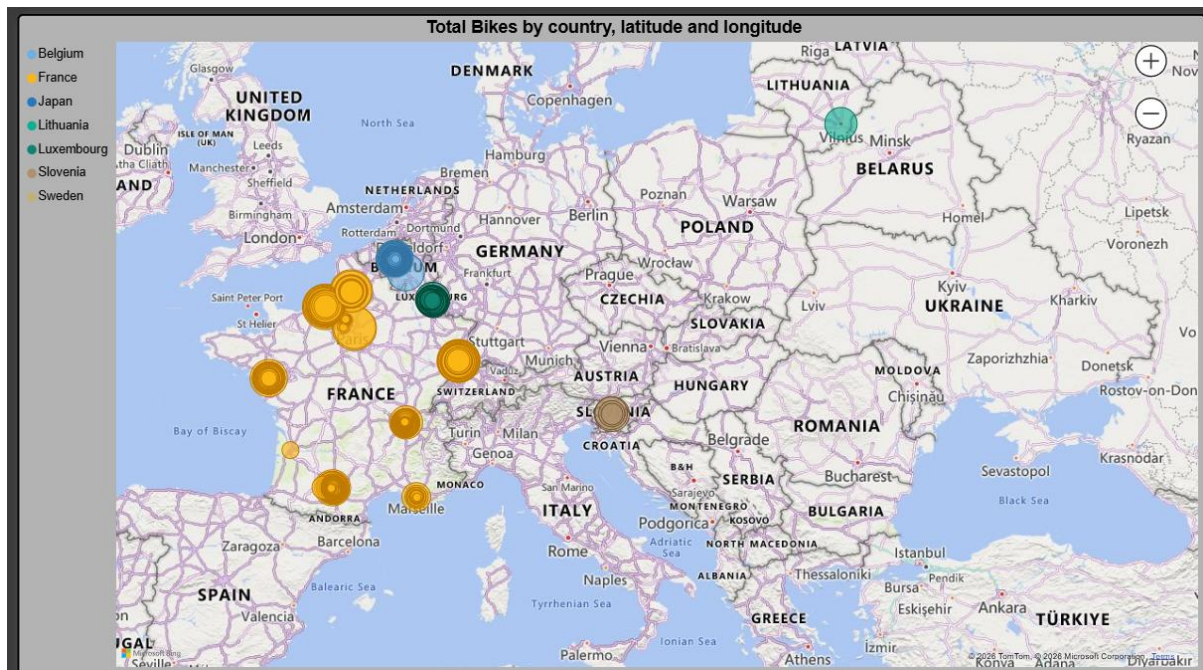
Total Capacity = SUM(Fact_Bike_status[bike stands])

Utilization % = DIVIDE([Total Bikes], [Total Capacity])

8. Analysis and Visualizations (Power BI)

SCREENSHOT OF DASHBOARD /REPORT





9. Insights & Conclusions

Key Findings:

The analysis reveals the following key performance indicators:

- Total Bikes Available: 13K
- Total Station Capacity: 40K
- Available Bike Stands: 22K
- Overall Utilization Rate: 32.76%

These metrics indicate that the current system has significant unused capacity, suggesting opportunities to improve bike distribution and station usage efficiency.

Insights:

The dashboard analysis highlights several important findings:

- **Low Utilization:** The overall utilization rate of approximately 33% indicates that a large portion of station capacity remains unused. This suggests inefficiencies in bike allocation or uneven demand across locations.
- **Geographic Concentration:** The majority of bikes are located in France, which accounts for the largest share of the network's bike availability. Other countries contribute relatively smaller portions of the system.
- **Network Growth Trend:** The trend analysis shows a steady increase in total bikes over time, indicating expansion of the bike-sharing infrastructure.
- **High-Performance Stations:** Several stations such as Balzac Boulevard and Magellan maintain significantly higher bike counts, suggesting that these locations function as major operational hubs.
- **Regional Utilization Differences:** Some countries exhibit higher proportions of available bike stands compared to bikes, indicating lower demand or inefficient distribution.

10. Conclusions

The Bike Station Availability & Utilization Dashboard demonstrates how data-driven insights can improve operational efficiency within bike-sharing networks. By identifying utilization gaps, high-demand stations, and geographic distribution patterns, organizations can make informed decisions to enhance service accessibility, optimize resource allocation, and support the sustainable growth of urban mobility solutions.